# Big Data in Vision Neuroscience & Health

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# 01

## What Makes Data Big?

Diebold, Francis. (2000). Big Data" Dynamic Factor Models for Macroeconomic Measurement and Forecasting.
https://www.researchgate.net/publication/2554936_Big_Data_Dynamic_Factor_Models_for_Macroeconomic_Measurement_and_Forecasting

# What Makes Data "Big"?

The "Four V's" framework defines the unique challenges and opportunities of massive scientific datasets.

## Volume

From gigabytes to petabytes, driving new infrastructure needs.

## Variety

Structured tables vs. unstructured images, notes, and signals.

## Velocity

Real-time data streams requiring rapid analysis and decisions.

## Veracity

Ensuring data quality, accuracy, and reliability for valid conclusions.

"Six V's" adds **Variability** and **Value**, and recently adding **Visualization**.

# The Volume Explosion

Advancements in acquisition tech drive an exponential growth in data **volume** and **speed** of production, moving from gigabytes to petabytes, necessitating significant computational resources and specialized software.

### High-Resolution 7T fMRI

Single scan generates several gigabytes of data. E. g. **NatPAC**: Minecraft gameplay, eye-tracking, and 7T fMRI.

### Natural Scenes Dataset (NSD)

Comprises over 73,000 fMRI scans from just eight subjects viewing 10,000 images.

### MIMIC-IV Database

Contains vast amounts of clinical data from ICU patients, includes vast amounts of clinical data, physiological waveforms, and notes.

# Variety & Veracity: The Hidden Hurdles

## Variety: Data Formats

Data comes in structured and unstructured formats, the latter being far more challenging to process.

- Structured: Patient demographics, lab results, ICD-10 codes.
- Unstructured (80-90%): Clinical notes, DICOM images, EEG traces.

## Veracity: Data Quality

Ensuring data quality and reliability is paramount, as errors and noise can significantly impact findings.

- Neuroimaging: Motion artifacts, scanner noise, protocol inconsistencies.
- Health Records: Missing entries, incorrect coding, NLP label errors.

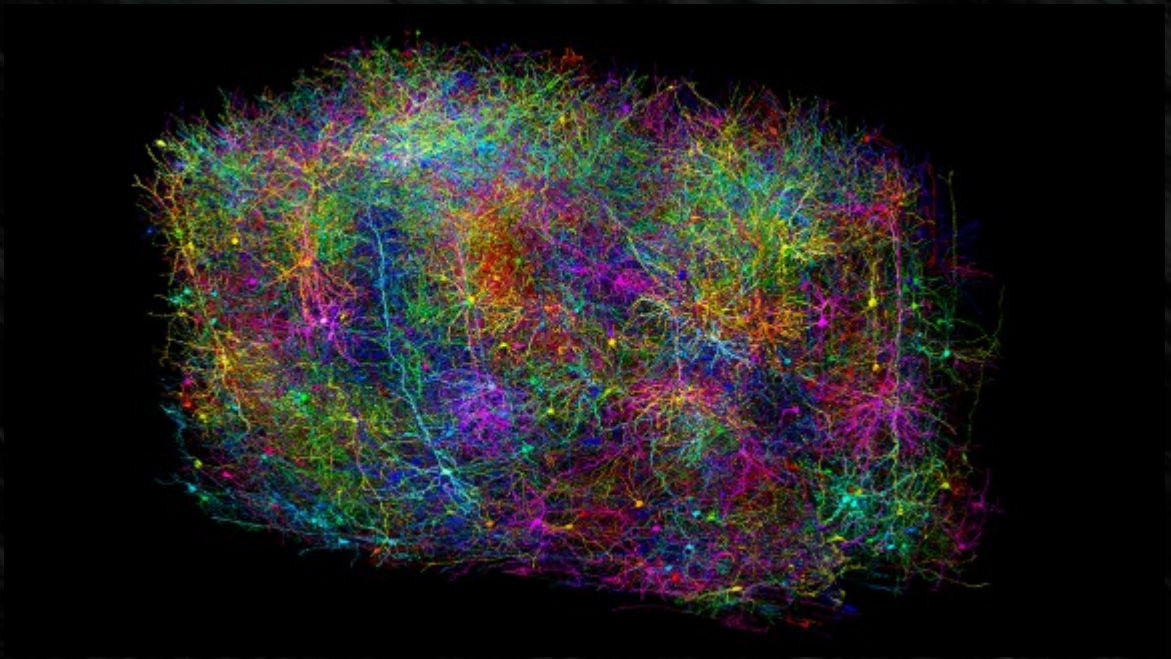
https://www.nature.com/collections/bdigiaicbd

https://www.nature.com/immersive/d42859-025-00001-w/index.html

# The MICrONS Project

## Dense reconstruction of functional and anatomical information in sensory brain

The MICrONS Project set out to create a massively dense reconstruction of the structural connections and functions of an entire millimeter volume of mouse visual cortex, along with new artificial intelligence-driven tools. The package of papers includes the description of the publicly-available resource and an initial compendium of papers using the resource to explore and model visual function and computation.



# Example of Lapan dataset

## Sources of data

- Erg – Electroretinography
- Fdt – Frequency doubling technology
- Medical records
- Right eye – Eye Tracking
- Survey – Questionnaires

## Types of data

- Quantitative / aggregates
- Temporal numeric
- Qualitative: categorical, binominal
- Spatial coordinates, displacements
- Image

# Example of Lapan dataset

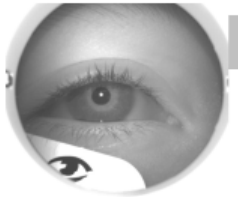
**RET*eval*™**

Número de série: R002055

Protocolo de teste: Flash/cintilação ISCEV Fotópica e PnNR Td

0% 95% 97.5% 100%

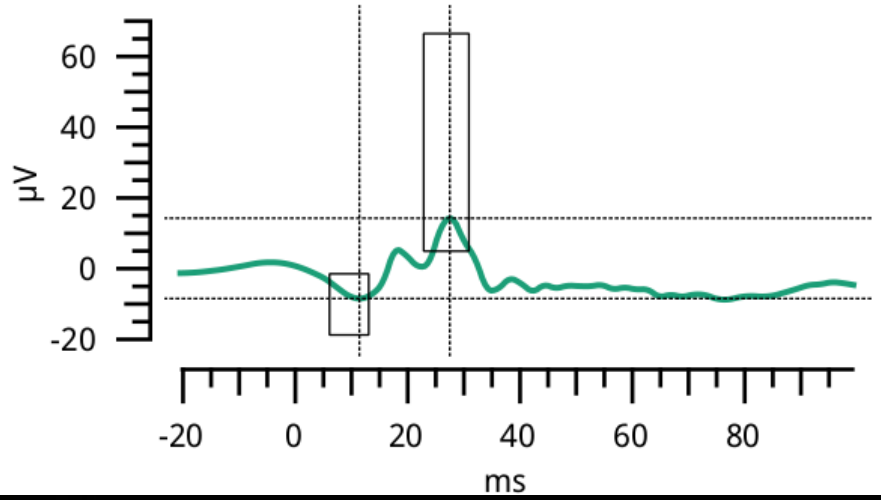
Tempo



**Teste #1: Flash: 85 Td·s, Cromaticidade (0,33, 0,33)**

## Olho direito (Pupila 3,0 mm)

| curva-a | | curva-b | |
|---|---|---|---|
| ms | µV | ms | µV |
| 6,1 ↔ 13,1 | -1,5 ↔ -18,7 | 22,8 ↔ 30,9 | 13,4 ↔ 74,9 |
| 11,4 (55%) | -8,5 (72%) | 27,5 (33%) | 22,7 (20%) |




| ID | Test_ID | Test_Date | Test_Time | Test_Meth | Patient_ID | Age | Dx | Notes | ODCT | ODN1 | ODN2 | ODN3 | ODN4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 11 | 18252.252 | ### | 15:22 | 4 | 10000 | 36 | | | 36,228001 | 36,228001 | 35,040001 | 33,416 | 32,877003 |
| 13 | 18254.252 | ### | 16:34 | 4 | 168584 | 17 | | SC. | 29,931002 | 29,443001 | 35,496002 | 35,265003 | 39,224003 |
| 15 | 18256.252 | ### | 10:47 | 4 | 168610 | 29 | | | 33,795002 | 36,228001 | 36,752003 | 36,228001 | 33,416 |
| 17 | 18258.252 | ### | 11:09 | 4 | 160554 | 7 | | COM FILTR | 24,684002 | 25,484001 | 26,279001 | 31,283001 | 24,410002 |
| 18 | 18261.252 | ### | 14:16 | 4 | 168614 | 40 | | SC. | 25,005001 | 27,180002 | 28,005001 | 27,180002 | 29,921001 |
| 19 | 18263.252 | ### | 14:45 | 4 | 168621 | 7 | | | 30,995001 | 30,995001 | 24,684002 | 31,283001 | 33,416 |
| 21 | 18265.252 | ### | 15:03 | 4 | 168619 | 40 | | | 31,055002 | 23,901001 | 29,573002 | 22,538002 | 26,668001 |
| 23 | 18267.252 | ### | 15:35 | 4 | 168824 | 9 | | | 24,684002 | 18,039001 | 18,831001 | 18,548 | 30,315001 |

Em entrevista de anamnese mãe relatou que adolescente vem apresentando dificuldades de atenção, ansiedade (rói unhas, fala rapidamente e chegou a arrancar parte dos cabelos), irritabilidade e timidez excessiva. Além disso, apresenta baixa persistência (perde o interesse pelas atividades com facilidade), desorganização e agitação motora. Apresenta histórico de dificuldades escolares, apesar de nunca ter sido reprovado. Aos 08 anos de idade chegou a fazer tratamento psicológico para comprometimento de atenção. Não apresenta histórico de problemas no parto, nem histórico de doenças graves ou problemas neurológicos na infância. Em 02/2017 realizou Ressonância Magnética não tendo sido detectado nenhuma alteração

# 02

## Why Scale Matters

# Capturing Real-World Complexity

Large-scale datasets provide the statistical power and diversity to capture the complexity of the real world, overcoming the limitations of small, homogeneous samples.

## Representative Sampling

The UK Biobank (500k+ individuals) provides a more representative picture of human biology across ages, ethnicities, and health conditions.

## Ecological Validity

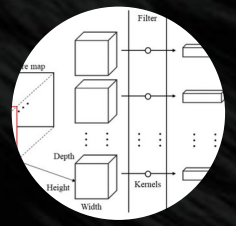
Datasets like the NSD capture the full range of real-world visual variability, leading to more robust and generalizable models.

# Enabling Complex Models & Generalization

The scale of a dataset is a critical factor in training powerful deep learning models, providing the necessary fuel to learn intricate patterns and achieve high generalization.

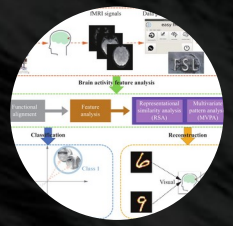
Massive Dataset
(e.g., NSD)

→




Complex AI Model
(e.g., Brain Diffuser)

→




Accurate Prediction
& Generalization

Large datasets allow models like the "Brain Diffuser" to learn detailed mappings between the visual world and neural code, enabling accurate prediction and reconstruction of perceived images.

# Facilitating New Discoveries

Large-scale datasets enable a powerful, data-driven approach to science, allowing researchers to identify novel patterns and generate new hypotheses that would be impossible with traditional methods.

### Drug Discovery

Mining EHR data to identify unexpected drug-drug interactions or repurpose existing drugs for new indications.

### Connectome Mapping

Analyzing DTI and microscopy data to build detailed maps of brain wiring and reveal new principles of organization.

### In Silico Experimentation

Rapidly testing hypotheses on large-scale brain models to explore disease effects or mechanisms of learning.

# 03

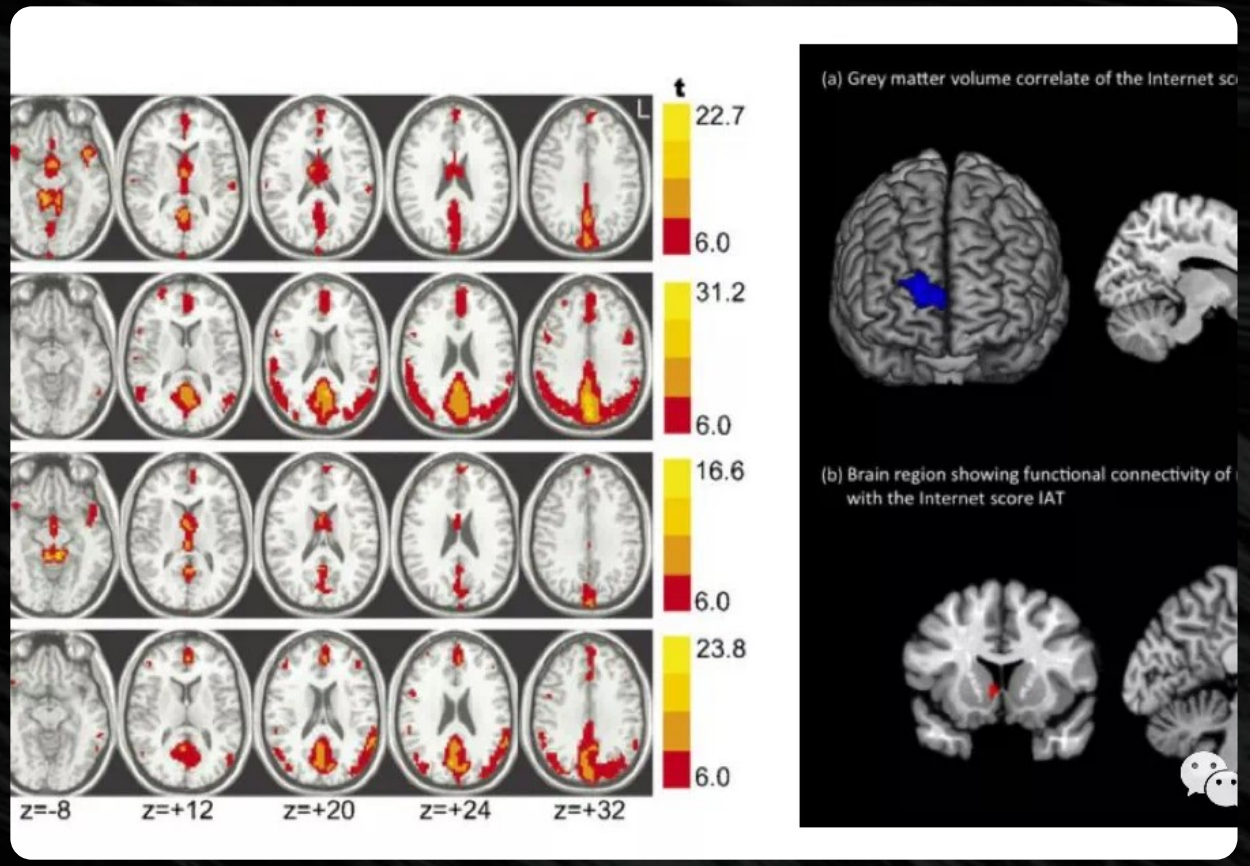
## Proeminent Datasets

https://naturalscenesdataset.org/

# Natural Scenes (NSD)

A landmark resource providing an unprecedented scale of high-quality 7T fMRI data to study how the brain processes naturalistic visual information.

- 8 Subjects: Densely sampled to allow reliable individual-level analysis.
- 73,000+ fMRI Scans: A massive collection for training complex models.
- 10,000 Unique Images: Shown multiple times to each subject for reliability.
- Eye-Tracking & Annotations: Rich, multimodal data for comprehensive analysis.



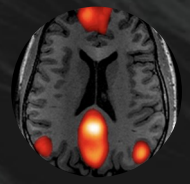
# THINGS-data

A massive, multimodal resource for investigating the neural and behavioral foundations of object recognition at scale.

## Stimuli

1,854 object concepts & 26k images

+



## Neuroimaging

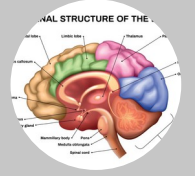
fMRI (8.7k images) & MEG (22.4k images)

+

## Behavior

4.7M similarity judgments from 12k+ participants

=



## Spatiotemporal Map

A holistic view of object processing

# Other Landmark Datasets

## UK Biobank

A monumental resource for multimodal health data, linking genetics to lifestyle and disease.

- 500,000+ Participants: A massive, diverse cohort.
- Genomic & Imaging Data: Links genes to brain structure.

## ENIGMA Consortium

A global alliance demonstrating the power of meta-analysis for neuroimaging genetics.

- 1,400+ Scientists: Across 43 countries.
- Standardized Pipelines: Ensuring reproducible results.

# 04

## Data Types & Tools

# A Spectrum of Data Types

### Neuroimaging
fMRI, MRI, PET

### Electrophysiology
EEG, MEG, Spike Trains

### Behavioral & Cognitive
Reaction times, accuracy scores

### Clinical & EHR
Diagnoses, medications, clinical notes

### Genomic & Omics
Genomics, Transcriptomics

### Integration Challenge
Combining modalities for holistic insights

# The Python Analytic Stack

A powerful and flexible ecosystem of libraries forms the foundation for modern data analysis in neuroscience and health.

## Core Data Science

NumPy & Pandas: For data manipulation and analysis.
Scikit-learn: For machine learning algorithms.

## Specialized Libraries

Nilearn & BrainIAK: For neuroimaging analysis.

PyDicom & MIMIC-Extract: For medical data processing.

# Decoding Visual Semantics from fMRI

A machine learning pipeline to predict semantic features of images from brain activity.

## 1. Load Data

Extract fMRI time-series from V1 ROI.

→

## 2. Train Model

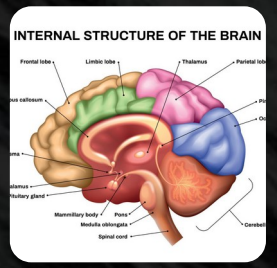
Use Ridge Regression to predict CLIP image features.

→

## 3. Evaluate

Assess performance using correlation.

→

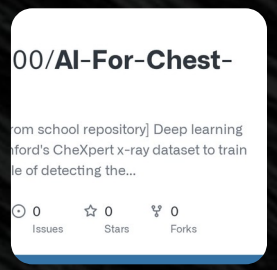



## 4. Interpret

Analyze model weights to understand voxel coding.

# Classifying Pathologies in Chest X-rays

Building a deep learning model for automated pathology detection using the NIH Chest X-ray Dataset.

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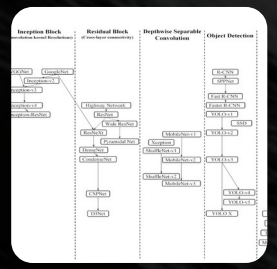


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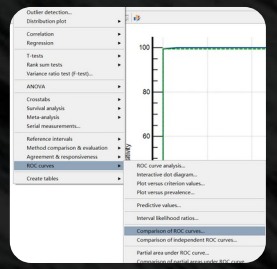



## 1. Load & Visualize

Use libraries like PyDicom to load images.

## 2. Preprocess & Augment

Resize, normalize, and augment data.

## 3. Build & Train CNN

Use a Convolutional Neural Network.

## 4. Evaluate & Diagnose

Monitor AUC and address class imbalance.

# 05

## Challenges & Ethics

# Methodological Challenges

## Data Quality & Bias

"Garbage in, garbage out" is a major risk. Selection bias can lead to models that fail to generalize.

## Curse of Dimensionality

High-dimensional data (e.g., genomics) can be sparse, leading to overfitting where models learn noise, not signal.

## Reproducibility

Complex analysis pipelines make reproducibility a challenge. Data and code sharing are essential but difficult.

# Methodological Challenges

## Common Data Quality Issues

- Missing values and incomplete records
- Inconsistent coding across systems
- Measurement errors and artifacts
- Temporal inconsistencies

## Types of Bias

- Selection bias in sampling
- Confirmation bias in analysis
- Demographic underrepresentation
- Temporal bias in longitudinal studies

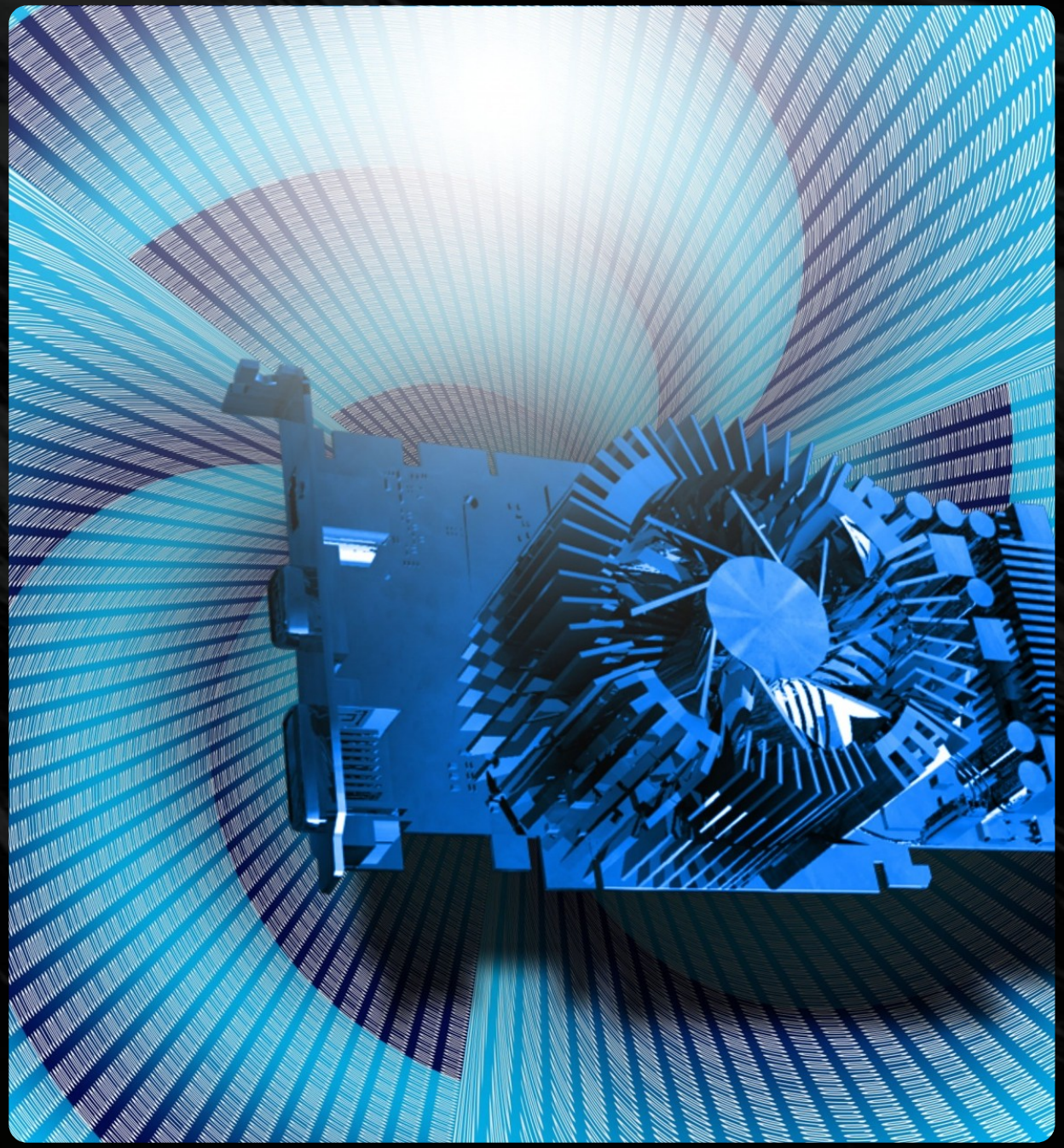
# Reproducibility and Data Sharing

## Reproducibility Best Practices

- Version control for code and data
- Containerized environments
- Comprehensive documentation
- Automated testing pipelines

## Data Sharing Challenges

- Privacy and security concerns
- Storage and bandwidth limitations
- Legal and ethical restrictions
- Standardization requirements



# Ethics: Privacy & Consent

The use of sensitive personal data in neuroscience and health research demands robust ethical frameworks to protect participant rights and maintain public trust.

## Informed Consent

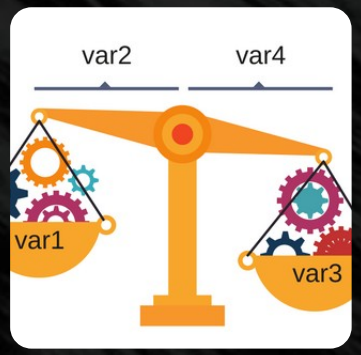
Must be clear about long-term data use and sharing, especially with complex data like brain scans.

## Data Privacy & Security

Neuroimaging can reveal mental states. De-identification is not foolproof, requiring secure storage and new techniques like federated learning.

# Ethics: Algorithmic Fairness

AI models trained on biased data can perpetuate or amplify health disparities, leading to unfair and discriminatory outcomes for marginalized groups.




→

→

### Biased Training Data
Over-representation of a demographic group.

### Unfair AI Outcomes
Disparities in diagnosis or treatment.

### Fairness & Equity
The goal for responsible AI.

Solutions include using diverse training data, developing fair algorithms, and maintaining a human-in-the-loop for review.

# 06

## Future Directions

# The Future: Global Collaborations

The future of big data lies in massive, multi-institutional consortia that pool resources and data to tackle challenges beyond the scope of any single lab.

### TCGA & APOLLO Network

Landmark cancer research integrating genomics and proteomics with clinical data.

### BICAN

The BRAIN Initiative's effort to map the human brain's 200 billion cells with a commitment to open science.

https://alleninstitute.org/news/global-effort-to-map-the-human-brain-releases-first-data/

https://portal.brain-map.org/

# BRAIN Initiative® Cell Atlas Network (BICAN)

**Mouse**

- Allen Brain Cell Atlas
- MapMyCells
- Cell Type Knowledge Explorer
- Patch-Seq Datasets
- Mouse Brain Atlas
- Mouse Brain Connectivity Atlas
- Brain Observatory
- Cell Type Database
- Developing Mouse Brain Atlas
- Mouse Spinal Cord Atlas
- Gene Expression Navigator Adult Mouse
- Gene Expression Navigator Developing Mouse
- Sleep Study

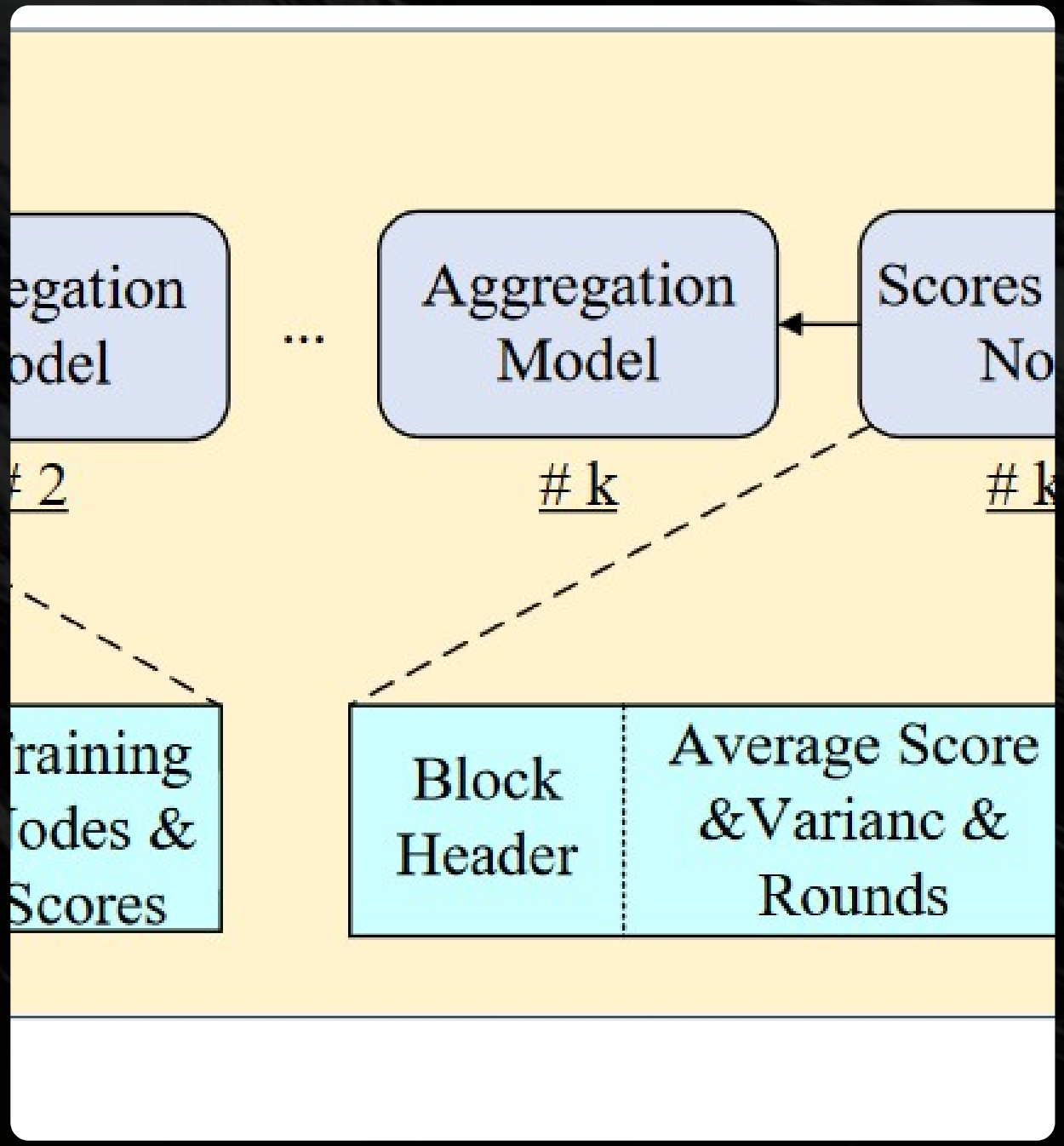
**Human**

- Seattle Alzheimer's Disease Brain Cell Atlas
- Allen Brain Cell Atlas
- MapMyCells
- Cell Type Knowledge Explorer
- Patch-Seq Datasets
- Cell Type Database
- Human Brain Atlas
- Developing Human Brain Atlas
- Aging, Dementia, and TBI Study
- Ivy Glioblastoma Atlas Project
- Schizophrenia Study
- Autism Study

**Non-Human Primate**

- Cell Type Knowledge Explorer
- Developing Non-Human Primate Atlas

The Brain Modeling Toolkit (BMTK) is an open-source software package for modeling and simulating large-scale neural network models. It supports a range of modeling resolutions, including multi-compartment, biophysically detailed models, point-neuron models, and population-level firing rate models.




# The Future: Privacy-Preserving Tech

New technologies are emerging to unlock the potential of distributed, sensitive datasets while upholding the highest standards of data privacy and ethical responsibility.

### Federated Learning

Trains models across decentralized data holders (e.g., hospitals) without sharing the raw data itself.

### Other Techniques

Methods like differential privacy and secure multi-party computation add layers of protection for sensitive data.

# Secure analytics platform for NHS electronic health records

## Data Sources

- Overview
- SystmOne primary care
- EMIS primary care
- Covid-19 test results
- Covid-19 therapeutics
- Emergency attendances
- High Cost Drugs
- Hospital admissions
- In-hospital deaths (covid-19 only)
- Intensive care admissions (covid-19 only)
- International Severe Acute Respiratory and emerging Infection Consortium
- Registered deaths
- UK Renal Registry

British Journal of General Practice
bringing research to clinical practice

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Research

### Trends and clinical characteristics of COVID-19 vaccine recipients: a federated analysis of 57.9 million patients' primary care records *in situ* using OpenSAFELY

Helen J Curtis, Peter Inglesby, Caroline E Morton, Brian MacKenna, Amelia Green, William Hulme, Alex J Walker, Jessica Morley, Amir Mehrkar, Seb Bacon, George Hickman, Chris Bates, Richard Croker, David Evans, Tom Ward, Jonathan Cockburn, Simon Davy, Krishnan Bhaskaran, Anna Schultze, Christopher T Rentsch, Elizabeth J Williamson, Anna Rowan, Louis Fisher, Helen I McDonald, Laurie Tomlinson, Rohini Mathur, Henry Drysdale, Rosalind M Eggo, Kevin Wing, Angel YS Wong, Harriet Forbes, John Parry, Frank Hester, Sam Harper, Shaun O'Hanlon, Alex Eavis, Richard Jarvis, Dima Avramov, Paul Griffiths, Aaron Fowles, Nasreen Parkes, Ian J Douglas, Stephen JW Evans, Liam Smeeth, Ben Goldacre and (The OpenSAFELY Collaborative)

British Journal of General Practice 2022; 72 (714): e51-e62. **DOI:** https://doi.org/10.3399/BJGP.2021.0376

https://www.princeton.edu/news/2020/03/31/multi-year-datasets-suggest-projecting-outcomes-peoples-lives-ai-isnt-so-simple

# Lesson: The Fragile Families Challenge

A landmark study that serves as a cautionary tale about the limits of predictive modeling, even with massive, high-quality datasets.

| 160 | 13,000+ | ↓ |
|---|---|---|
| Research Teams | Variables per Family | Low Accuracy |

Despite the scale and collective expertise, models failed to accurately predict life outcomes. This highlights the dangers of over-relying on predictive models for high-stakes decisions and underscores the need for scientific humility and better theory.

# Thank you.